BEAM DESIGN CALCULATION

Project Name	:	G+2 RB PLINTH BEAM DESIGN
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-1.981 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 03:00:45
Level Designed	:	1.981 m

Sr.No.	Symbol		Definitions
1	α	=	Angle formed with horizontal by diagonal reinforcement
2	ε1	=	Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone
3	εm	=	Average steel strain at level considered
4	Acr	=	Distance from the point considered to the surface of the nearest longitudinal bar
5	Adr	=	Area of diagonal reinforcement in coupler beam
6	AstCalc	=	Area of Tension reinforcement required in sqmm
7	Ast comp	=	Ast at compression side
8	AstCrack	=	Area of Tension reinforcement for Crack Width required in sqmm
9	AstPrv	=	Area of reinforcement provided at given section in sqmm
10	Asv	=	Area of shear reinforcement required per meter length in sqmm
11	Asv Reqd	=	Min area shear reinforcement as per clause 41.4.3
12	Asv Total Prv	=	Total area of shear reinforcement provided, including that for torsion
13	Asv Torsion	=	Area of Stirrups over the corner bars to resist shear and torsion as per clause 41.4.3
14	Asv Torsion Prv	=	Area of shear reinforcement resisting torsion (outer legs only)
15	b	=	Width of the Beam in mm
16	bw	=	Width of Web in mm
17	Ceff	=	Effective Cover of Beam in mm
18	Cmin	=	Clear cover to top and bottom reinforcement in mm
19	Cside	=	Clear cover to side reinforcement in mm
20	d	=	Effective depth of Beam in mm
21	d'	=	Clear cover to reinforcement at compression side.
22	D	=	Depth of Beam in mm
23	D'	=	Distance (c/c) between top-most and bottom-most layer
24	Df	=	Thickness of Flange in mm
25	Ec	=	Modulus of elasticity of concrete
26	Es	=	Modulus of elasticity of steel
27	FcPerm	=	Permissible Stress in Concrete required in N/sqmm

28	Fst	=	Stress in steel
29	FstPerm	=	Permissible Stress in Steel required in N/sqmm
30	Legs	=	Number of legs of the shear reinforcement
31	lo	=	Effective Length of Beam in mm
32	Mh	=	Hogging Moment of Resistance in kNm
33	Ms	=	Sagging Moment of Resistance in kNm
34	M_{Tu}	=	Converted Bending Moment due to Torsion at a section in kNm
35	Mu	=	Factored Bending Moment in Limit State at a section in kNm
36	Mud	=	Total Design Moment = Mu + M_{Tu} in kNm
37	MuLim	=	Limiting Moment of Resistence of Singly Reinforced Section
38	NA angle (φ)	=	It is a neutral axis angle corresponding to load angle to find out MCap
39	Pcclc	=	Calculated percentage compression steel
40	Ptclc	=	Calculated percentage tension steel
41	Ptmin	=	Minimum percentage steel as per code
42	PtPrv	=	provided percentage steel
43	R	=	Moment Ratio Mud/Bd^2
44	Reinf	=	Rebar arrangement provided
45	sp	=	Spacing Between bars at outer most layer
46	Spc1 to Spc6	=	Max Stirrup Criteria in mm
47	Stirrup Dia	=	Diameter of shear stirrups in mm
48	SvCalc	=	Stirrup spacing calculated as per Asv in mm
49	SvPrv	=	Stirrup spacing provided in m
50	Tc	=	Permissible Shear stress in concrete at a given section in N/sqmm Calculated as per Tensile reinforcement
51	Tce	=	Earthquake inducted shear stress in N/sqmm
52	Tcep	=	Permissible value of shear stress for checking of coupling action in N/sqmm
53	Tu	=	Factored Torsional Moment in Limit State at a section in kNm
54	Tv	=	Shear stress due to Vud in N/sqmm
55	V^{D+L}	=	Shear force 1.2x(Dead Load + Live Load) in kN
56	Vc	=	Shear capacity of concrete section due to Tc in kN
57	Vu	=	Factored Shear Force in Limit State at a section in kN
58	Vu-A1	=	V^{D+L}Left - 1.4 x (MsLeft + MhRight)/L in kN
59	Vu-A2	=	V^{D+L}Left + 1.4 x (MhLeft + MsRight)/L in kN
60	Vu-B1	=	V^{D+L}Right + 1.4 x (MsLeft + MhRight)/L in kN
61	Vu-B2	=	V^{D+L}Right - 1.4 x (MhLeft + MsRight)/L in kN
62	Vud	=	Design Shear Force (Max of Vus and Vu-sway) in kN
63	Vue	=	Earthquake inducted shear in kN
64	Vu-sway	=	Max of (Vu-A1 and Vu-A2) & (Vu-B1 and Vu-B2) in kN
65	Vus	=	Vud - Vc Design Shear Force for reinforcement, in kN
66	Vut	=	Total Shear Force = Vu + V_{Tu} in kN

67	V_{Tu}	=	Converted Shear Force due to Torsion at a section in kN
68	Wcr	=	Surface Crack Width
69	WcrPerm	=	Permissible Crack Width required in mm
70	X1	=	Smaller c/c dimension of outermost Stirrups
71	Y1	=	Larger c/c dimension of outermost Stirrups
			All 6mm diameter bars are assumed 250 grade with fy = 250 N/sqmm

All Forces are in 'kN', 'kNm', Stress in 'N/sqmm' & Dimension are in 'mm'.

Code References

IS 456

Sr.No.	Item		Clause / Table
1.	Ptmax	:	26.5.1.1
2.	Ptmin	:	26.5.1.1
3.	M_{Tu}	:	41.4.2
4.	V_{Tu}	:	41.3.1
5.	Ptclc	:	38
6.	Tc	:	40.2.1
7.	Tcmax	:	40.2.3
8.	Asv	:	40.4
9.	Min Shear Reinf	:	26.5.1.6
10.	Max Stirrup Spacing	:	26.5.1.5
11.	Shear Reinf - Torsion	:	26.5.1.7
12.	Side Face Reinforcement	:	26.5.1.3
13.	crack width calculation	:	Annex-F
14.	cracking	:	Clause 35.3.2

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Sr.No.	Item		Clause / Table
1.	Ptmax	:	6.2.2
2.	Ptmin	:	6.2.1
3.	Astmin	:	6.2.3 & 6.2.4
4.	Svclc	:	6.3.5
5.	Shear Capacity	:	6.3.4
6.	Coupling Beams	:	10.5

Group	:	**G1**	
Beam No	:	**B1**	
Analysis Reference (Member)	:	1 (1.981 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm

Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	12	19	14	19	12
Critical L/C - RCDC	13	8	15	10	15	8
Mu (kNm)	58.87	47.76	31.43	72.46	28.34	79.36
Tu (kNm)	0.44	0.57	0	0	0	0.57
M_{Tu} (kNm)	0.75	0.95	0	0	0	0.95
Mud (kNm)	59.62	48.71	31.43	72.46	28.34	80.31
MuLim (kNm)	102	102	102	102	102	102
R	1.893	1.547	0.998	2.301	0.9	2.551
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.438	0.35	0.219	0.547	0.2	0.617
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	372.84	298.25	186.54	465.56	170.2	525.32
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
				Area of reinforcement required for

D = AstCrack = Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	58.76	46.14	69.7
Tu (kNm)	0	0.56	0.56
V_{Tu} (kN)	0	3.93	3.93
Vut (kN)	58.76	50.08	73.63
Tv (N/sqmm)	0.69	0.59	0.87
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	2.07	0	16.94
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	175.897	250.458
Asv Torsion Prv (sqmm)	-	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group	:	**G1**	
Beam No	:	**B2**	
Analysis Reference (Member)	:	2 (1.981 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	37.16	34.51	45.74	69.71	37.77	94.69
Tu (kNm)	0	0.48	0.48	0.64	0.17	0.17
M_{Tu} (kNm)	0	0.81	0.81	1.08	0.28	0.28
Mud (kNm)	37.16	35.32	46.55	70.79	38.05	94.97
MuLim (kNm)	102	102	102	102	102	102
R	1.18	1.122	1.478	2.248	1.208	3.016
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.262	0.248	0.334	0.533	0.268	0.757
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	222.82	211.05	283.86	453.19	228.48	644.27
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

Maximum of % reinforcement required for

B'	=	Ptclc	=	Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	61.64	43.31	66.87
Tu (kNm)	0.64	0.17	0.17
V_{Tu} (kN)	4.46	1.16	1.16
Vut (kN)	66.1	44.47	68.03
Tv (N/sqmm)	0.78	0.52	0.8
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	9.41	0	11.34
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	228.919	145.906	220.467
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm

Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G1**	
Beam No	:	**B3**	
Analysis Reference (Member)	:	194 (1.981 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	15	15	15	15	15	15
Critical L/C - RCDC	11	11	11	11	11	11
Mu (kNm)	-	-	-	34.77	25.21	8.73
Tu (kNm)	12.35	12.35	12.35	12.35	12.35	12.35
M_{Tu} (kNm)	20.85	20.85	20.85	20.85	20.85	20.85
Mud (kNm)	20.85	20.85	20.85	55.63	46.06	29.59
MuLim (kNm)	102	102	102	102	102	102
R	0.662	0.662	0.662	1.767	1.463	0.94
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.406	0.33	0.206
Pcclc (%) (C)	0.143	0.143	0.143	0	0	0.082
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	345.15	280.65	175.08
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	15	15	15
Critical L/C - RCDC	11	11	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	29.17	25.74	22.31
Tu (kNm)	12.35	12.35	12.35
V_{Tu} (kN)	85.94	85.94	85.94
Vut (kN)	115.11	111.68	108.25
Tv (N/sqmm)	1.35	1.31	1.27
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	58.42	54.99	51.56
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	744.088	733.227	722.365
Asv Torsion Prv (sqmm)	744.74	744.74	744.74
Asv Reqd (sqmm/m)	437.37	411.69	386
SvCalc (mm)	135	135	135
SvPrv (mm)	135	135	135
Asv Total Prv (sqmm)	744.74	744.74	744.74

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G2**
Beam No : **B4**
Analysis Reference (Member) : 21 (1.981 m)
Beam Length : 3924 mm
Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456
Beam Type : Regular Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	12	19	14	19	12
Critical L/C - RCDC	13	8	15	10	15	8
Mu (kNm)	69.36	47.07	44.36	70.91	34.01	70.63
Tu (kNm)	0.21	0.22	0.2	0.2	0.2	0.22
M_{Tu} (kNm)	0.36	0.37	0.34	0.33	0.34	0.37
Mud (kNm)	69.72	47.44	44.7	71.25	34.35	71
MuLim (kNm)	102	102	102	102	102	102
R	2.214	1.507	1.42	2.263	1.091	2.255
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.523	0.34	0.319	0.537	0.241	0.534

Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	445.36	289.76	271.63	456.58	204.92	454.74
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	45.37	41.47	53.25
Tu (kNm)	0.2	0.22	0.22
V_{Tu} (kN)	1.37	1.52	1.52
Vut (kN)	46.75	42.99	54.77
Tv (N/sqmm)	0.55	0.51	0.64
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	154.025	142.806	180.086
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G2**	
Beam No	:	**B5**	
Analysis Reference (Member)	:	22 (1.981 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	51.22	31.46	50.49	58.71	50.96	101.73
Tu (kNm)	0.18	0.21	0.21	0.21	0.17	0.17
M_{Tu} (kNm)	0.3	0.35	0.35	0.36	0.29	0.29
Mud (kNm)	51.52	31.81	50.84	59.08	51.25	102.01
MuLim (kNm)	102	102	102	102	102	102
R	1.636	1.01	1.615	1.876	1.628	3.24
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2

Ptclc (%) (B')	0.373	0.222	0.367	0.434	0.371	0.829
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	317.14	188.92	312.55	369.07	315.3	705.52
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	14	12	7
Critical L/C - RCDC	10	8	3
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	42.24	44.4	56.34
Tu (kNm)	0.22	0.17	0.13
V_{Tu} (kN)	1.5	1.18	0.88
Vut (kN)	43.73	45.58	57.22
Tv (N/sqmm)	0.51	0.54	0.67
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0.53
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	145.037	149.48	185.02
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G2**	
Beam No	:	**B6**	
Analysis Reference (Member)	:	195 (1.981 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	13	13	13	5	8	13
Critical L/C - RCDC	9	9	9	1	4	9
Mu (kNm)	-	-	3.02	75.59	44.77	11.4

Tu (kNm)	10	10	10	1.46	8.42	10
M_{Tu} (kNm)	16.88	16.88	16.88	2.47	14.21	16.88
Mud (kNm)	16.88	16.88	19.9	78.05	58.98	28.28
MuLim (kNm)	102	102	102	102	102	102
R	0.536	0.536	0.632	2.479	1.873	0.898
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.597	0.433	0.2
Pcclc (%) (C)	0.115	0.115	0.094	0	0	0.037
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	507.89	368.41	170.2
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	59.69	58.86	56.28
Tu (kNm)	1.46	1.46	1.46
V_{Tu} (kN)	10.16	10.16	10.16
Vut (kN)	69.85	69.02	66.44
Tv (N/sqmm)	0.82	0.81	0.78
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	13.16	12.33	9.75
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	266.006	263.386	255.213
Asv Torsion Prv (sqmm)	718.14	718.14	718.14

Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G3**	
Beam No	:	**B7**	
Analysis Reference (Member)	:	41 (1.981 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	12	19	14	19	12
Critical L/C - RCDC	13	8	15	10	15	8
Mu (kNm)	74.15	48.67	51.32	75	35.46	79.11
Tu (kNm)	0.13	0	0.28	0.33	0.28	0
M_{Tu} (kNm)	0.22	0	0.47	0.55	0.47	0
Mud (kNm)	74.37	48.67	51.79	75.56	35.94	79.11
MuLim (kNm)	102	102	102	102	102	102
R	2.362	1.546	1.645	2.4	1.141	2.512
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.564	0.35	0.375	0.574	0.253	0.606
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	479.9	297.96	318.98	488.85	214.98	516
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	9	12	7
Critical L/C - RCDC	5	8	3
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	48.24	44.9	58.68
Tu (kNm)	0.27	0	0
V_{Tu} (kN)	1.86	0	0
Vut (kN)	50.1	44.9	58.69
Tv (N/sqmm)	0.59	0.53	0.69
Tc (N/sqmm)	0.67	0.67	0.67

Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	2
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	166.78	-	-
Asv Torsion Prv (sqmm)	365.6	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G3**	
Beam No	:	**B8**	
Analysis Reference (Member)	:	42 (1.981 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design						
	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	58.89	34.17	56.66	66.38	50.75	104
Tu (kNm)	0.29	0.12	0.12	0	0.34	0.34
M_{Tu} (kNm)	0.49	0.2	0.2	0	0.58	0.58
Mud (kNm)	59.38	34.37	56.87	66.37	51.33	104.58
MuLim (kNm)	102	102	102	102	102	102
R	1.886	1.092	1.806	2.108	1.63	3.321
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.436	0.241	0.416	0.495	0.371	0.851
Pcclc (%) (C)	0	0	0	0	0	0.02
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	371.16	205.06	353.7	420.94	315.85	723.86
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	Mid	Right
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	46.01	46.93	58.71
Tu (kNm)	0	0.34	0.34
V_{Tu} (kN)	0	2.4	2.4
Vut (kN)	46.01	49.33	61.11

Tv (N/sqmm)	0.54	0.58	0.72
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	4.42
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	166.75	204.03
Asv Torsion Prv (sqmm)	-	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group : **G3**

Beam No : **B9**

Analysis Reference (Member) : 196 (1.981 m)

Beam Length : 1372 mm

Breadth (B) : 230 mm

Depth (D) : 430 mm

Effective Depth (d) : 370 mm

Design Code : IS 456

Beam Type : Cantilever Beam

Grade Of Concrete (Fck) : M25 N/sqmm

Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	13	13	13	5	8	13
Critical L/C - RCDC	9	9	9	1	4	9
Mu (kNm)	-	-	-	63.47	35.65	11.06
Tu (kNm)	11.5	11.5	11.5	3.19	9.98	11.5
M_{Tu} (kNm)	19.42	19.42	19.42	5.39	16.84	19.42
Mud (kNm)	19.42	19.42	19.42	68.86	52.49	30.48
MuLim (kNm)	102	102	102	102	102	102
R	0.617	0.617	0.617	2.187	1.667	0.968
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.516	0.38	0.212
Pcclc (%) (C)	0.133	0.133	0.133	0	0	0.056
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	438.98	323.71	180.66
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5

Critical L/C - RCDC	1	1	1
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	48.46	47.64	45.06
Tu (kNm)	3.19	3.19	3.19
V_{Tu} (kN)	22.19	22.19	22.19
Vut (kN)	70.66	69.83	67.25
Tv (N/sqmm)	0.83	0.82	0.79
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	13.97	13.14	10.56
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	321.718	319.098	310.925
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	: **G4**	
Beam No	: **B10**	
Analysis Reference (Member)	: 61 (1.981 m)	
Beam Length	: 3924	mm

Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	12	19	14	19	12
Critical L/C - RCDC	13	8	15	10	15	8
Mu (kNm)	70.28	53.04	44.86	82.57	32.82	93.44
Tu (kNm)	0	0	0.33	0.43	0.33	0
M_{Tu} (kNm)	0	0	0.56	0.72	0.56	0
Mud (kNm)	70.28	53.04	45.42	83.3	33.38	93.44
MuLim (kNm)	102	102	102	102	102	102
R	2.232	1.684	1.442	2.645	1.06	2.968
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.528	0.385	0.325	0.645	0.234	0.742
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	449.44	327.41	276.37	548.72	198.8	631.31
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required

D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	64.72	52.72	76.28
Tu (kNm)	0.43	0	0
V_{Tu} (kN)	2.99	0	0
Vut (kN)	67.7	52.72	76.28
Tv (N/sqmm)	0.8	0.62	0.9
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	11.01	0	19.59
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	227.498	-	-
Asv Torsion Prv (sqmm)	718.14	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group	:	**G4**	
Beam No	:	**B11**	
Analysis Reference (Member)	:	62 (1.981 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	51	35.51	50.79	79.94	44.06	107.99
Tu (kNm)	0.11	0	0	0	0.15	0.15
M_{Tu} (kNm)	0.19	0	0	0	0.26	0.26
Mud (kNm)	51.19	35.51	50.79	79.94	44.32	108.25
MuLim (kNm)	102	102	102	102	102	102
R	1.626	1.128	1.613	2.539	1.407	3.438
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.37	0.249	0.367	0.614	0.316	0.88
Pcclc (%) (C)	0	0	0	0	0	0.054
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	314.91	212.28	312.21	522.47	269.13	748.59
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement

B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	65.01	50.46	74.02
Tu (kNm)	0	0.15	0.15
V_{Tu} (kN)	0	1.06	1.06
Vut (kN)	65.01	51.52	75.07
Tv (N/sqmm)	0.76	0.61	0.88
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	8.32	0	18.38
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	167.74	242.301
Asv Torsion Prv (sqmm)	-	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	140
SvPrv (mm)	275	275	140
Asv Total Prv (sqmm)	365.6	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm
Side Face Reinforcement not required.		

Group	:	**G4**	
Beam No	:	**B12**	
Analysis Reference (Member)	:	197 (1.981 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	15	15	13	8	8	15
Critical L/C - RCDC	11	11	9	4	4	11
Mu (kNm)	-	-	6.94	59.48	42.1	7.59
Tu (kNm)	6.26	6.26	5.15	3.93	3.93	6.26
M_{Tu} (kNm)	10.56	10.56	8.69	6.63	6.63	10.56
Mud (kNm)	10.56	10.56	15.63	66.11	48.74	18.15
MuLim (kNm)	102	102	102	102	102	102
R	0.335	0.335	0.496	2.1	1.548	0.577
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.492	0.351	0.2
Pcclc (%) (C)	0.071	0.071	0.012	0	0	0.02
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	419.03	298.41	170.2
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	8	8	8
Critical L/C - RCDC	4	4	4
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	50.34	50.06	47.65
Tu (kNm)	3.93	3.93	3.93
V_{Tu} (kN)	27.33	27.33	27.33
Vut (kN)	77.67	77.39	74.98
Tv (N/sqmm)	0.91	0.91	0.88
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	20.98	20.7	18.29
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	366.595	365.715	358.087
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G5**
Beam No : **B13**
Analysis Reference (Member) : 191 (1.981 m)
Beam Length : 3924 mm
Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456
Beam Type : Regular Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	7	5	5	9	5	5
Critical L/C - RCDC	3	1	1	5	1	1
Mu (kNm)	14.51	17.13	9.25	7.47	-	9.56
Tu (kNm)	2.3	2.73	2.73	2.07	2.73	2.73
M_{Tu} (kNm)	3.88	4.6	4.6	3.49	4.6	4.6
Mud (kNm)	18.39	21.73	13.86	10.96	4.6	14.16
MuLim (kNm)	102	102	102	102	102	102
R	0.584	0.69	0.44	0.348	0.146	0.45
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2

Pcclc (%) (C)	0	0	0	0	0.031	0
PtPrv (%)	0.266	0.266	0.266	0.266	0.266	0.266
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement Provided	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	7	5
Critical L/C - RCDC	1	3	1
PtPrv (%)	0.266	0.266	0.266
Vu (kN)	19.59	6.97	23.58
Tu (kNm)	2.73	2.3	2.73
V_{Tu} (kN)	18.97	15.98	18.97
Vut (kN)	38.56	22.95	42.55
Tv (N/sqmm)	0.45	0.27	0.5
Tc (N/sqmm)	0.37	0.37	0.37
Vc (kN)	31.88	31.88	31.88
Vus=Vut-Vc (kN)	6.68	0	10.67
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	194.118	134.048	206.466
Asv Torsion Prv (sqmm)	718.14	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	140
SvPrv (mm)	140	275	140
Asv Total Prv (sqmm)	718.14	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G5**	
Beam No	:	**B14**	
Analysis Reference (Member)	:	193 (1.981 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	9	9	14	9	12	12
Critical L/C - RCDC	5	5	10	5	8	8
Mu (kNm)	5.1	11.15	8.38	9.7	10.8	27.46
Tu (kNm)	1.94	1.94	1.77	1.94	0.14	0.14

M_{Tu} (kNm)	3.28	3.28	2.99	3.28	0.24	0.24
Mud (kNm)	8.38	14.42	11.38	12.98	11.04	27.69
MuLim (kNm)	102	102	102	102	102	102
R	0.266	0.458	0.361	0.412	0.351	0.88
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.266	0.266	0.266	0.266	0.266	0.266
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement Provided	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	9	9	5
Critical L/C - RCDC	5	5	1
PtPrv (%)	0.266	0.266	0.266
Vu (kN)	16.52	7.78	24.02
Tu (kNm)	1.94	1.94	1.61
V_{Tu} (kN)	13.51	13.51	11.2
Vut (kN)	30.02	21.28	35.23
Tv (N/sqmm)	0.35	0.25	0.41
Tc (N/sqmm)	0.37	0.37	0.37
Vc (kN)	31.88	31.88	31.88
Vus=Vut-Vc (kN)	0	0	3.35
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	146.174	119.136	153.214
Asv Torsion Prv (sqmm)	365.6	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81

SvCalc (mm)	275	275	140
SvPrv (mm)	275	275	140
Asv Total Prv (sqmm)	365.6	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G5**	
Beam No	:	**B15**	
Analysis Reference (Member)	:	198 (1.981 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	7	7	7	12	7	7
Critical L/C - RCDC	3	3	3	8	3	3
Mu (kNm)	-	-	-	20.09	15	9.59
Tu (kNm)	9.88	9.88	9.88	9.42	9.88	9.88
M_{Tu} (kNm)	16.68	16.68	16.68	15.9	16.68	16.68
Mud (kNm)	16.68	16.68	16.68	36	31.68	26.27
MuLim (kNm)	102	102	102	102	102	102
R	0.53	0.53	0.53	1.143	1.006	0.834
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.253	0.221	0.2
Pcclc (%) (C)	0.114	0.114	0.114	0	0.011	0.048
PtPrv (%)	0.266	0.266	0.266	0.266	0.266	0.266
AstCalc (sqmm)	170.2	170.2	170.2	215.37	188.16	170.2
AstPrv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement Provided	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	12	12	12
Critical L/C - RCDC	8	8	8
PtPrv (%)	0.266	0.266	0.266
Vu (kN)	14.12	10.69	7.26
Tu (kNm)	9.42	9.42	9.42
V_{Tu} (kN)	65.55	65.55	65.55
Vut (kN)	79.67	76.24	72.81
Tv (N/sqmm)	0.94	0.9	0.86
Tc (N/sqmm)	0.37	0.37	0.37
Vc (kN)	31.88	31.88	31.88

Vus=Vut-Vc (kN)	47.79	44.36	40.93
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	505.075	494.456	483.838
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	357.75	332.06	306.38
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G6**	
Beam No	:	**B16**	
Analysis Reference (Member)	:	189 (1.981 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm

(Cmin)			
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	14	14	14	14	9	9
Critical L/C - RCDC	10	10	10	10	5	5
Mu (kNm)	-	-	-	10.92	40.55	58.13
Tu (kNm)	8.43	8.43	8.43	8.43	7.47	7.47
M_{Tu} (kNm)	14.23	14.23	14.23	14.23	12.62	12.62
Mud (kNm)	14.23	14.23	14.23	25.16	53.17	70.75
MuLim (kNm)	102	102	102	102	102	102
R	0.452	0.452	0.452	0.799	1.689	2.247
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.386	0.532
Pcclc (%) (C)	0.097	0.097	0.097	0.022	0	0
PtPrv (%)	0.266	0.266	0.266	0.709	0.709	0.709
AstCalc (sqmm)	170.2	170.2	170.2	170.2	328.3	452.89
AstPrv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement Provided	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	9	9	9
Critical L/C - RCDC	5	5	5
PtPrv (%)	0.709	0.709	0.709

Vu (kN)	22.9	29.66	33.64
Tu (kNm)	7.47	7.47	7.47
V_{Tu} (kN)	51.99	51.99	51.99
Vut (kN)	74.89	81.66	85.63
Tv (N/sqmm)	0.88	0.96	1.01
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85
Vus=Vut-Vc (kN)	27.04	33.81	37.78
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	451.359	472.541	484.969
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	282.82
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G6**	
Beam No	:	**B17**	
Analysis Reference (Member)	:	105 (1.981 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm

Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11
Mu (kNm)	24.31	25.55	26.86	82.57	23.79	67.3
Tu (kNm)	0.42	0.25	0	0.25	0.25	0.59
M_{Tu} (kNm)	0.72	0.42	0	0.42	0.42	1
Mud (kNm)	25.03	25.97	26.86	82.99	24.21	68.3
MuLim (kNm)	102	102	102	102	102	102
R	0.795	0.825	0.853	2.636	0.769	2.169
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.642	0.2	0.511
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.266	0.266	0.266	0.709	0.709	0.709
AstCalc (sqmm)	170.2	170.2	170.2	546.29	170.2	434.94
AstPrv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement Provided	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	0.709	0.709	0.709
Vu (kN)	62.43	35.45	57.09
Tu (kNm)	0.25	0.25	0.59
V_{Tu} (kN)	1.74	1.74	4.13
Vut (kN)	64.17	37.19	61.22
Tv (N/sqmm)	0.75	0.44	0.72
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85
Vus=Vut-Vc (kN)	16.32	0	13.37
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	208.092	123.638	208.819
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group : **G6**

Beam No	:	**B18**	
Analysis Reference (Member)	:	93 (1.981 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	15	13	13	13	18	15
Critical L/C - RCDC	11	9	9	9	14	11
Mu (kNm)	20.97	26.35	24.89	77.51	15.78	70.29
Tu (kNm)	0.15	0.16	0.16	0.16	0.16	0.15
M_{Tu} (kNm)	0.25	0.27	0.27	0.27	0.27	0.25
Mud (kNm)	21.21	26.62	25.16	77.79	16.05	70.53
MuLim (kNm)	102	102	102	102	102	102
R	0.674	0.846	0.799	2.47	0.51	2.24
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.594	0.2	0.53
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.266	0.266	0.266	0.709	0.709	0.709
AstCalc (sqmm)	170.2	170.2	170.2	505.84	170.2	451.32
AstPrv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement Provided	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

Maximum of % reinforcement required for

B'	=	Ptclc	=	Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	0.709	0.266	0.709
Vu (kN)	61.17	30.75	58.24
Tu (kNm)	0.16	0.16	0.15
V_{Tu} (kN)	1.13	1.13	1.02
Vut (kN)	62.3	31.88	59.25
Tv (N/sqmm)	0.73	0.37	0.7
Tc (N/sqmm)	0.56	0.37	0.56
Vc (kN)	47.85	31.88	47.85
Vus=Vut-Vc (kN)	14.45	0	11.4
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	199.667	104.477	189.696
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm

Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G6**	
Beam No	:	**B19**	
Analysis Reference (Member)	:	81 (1.981 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	13	13	18	15
Critical L/C - RCDC	16	9	9	9	14	11
Mu (kNm)	19.09	32.15	29.36	82.88	17.33	68.48
Tu (kNm)	0	0.55	0.55	0.55	0.41	0.15
M_{Tu} (kNm)	0	0.92	0.92	0.92	0.69	0.26
Mud (kNm)	19.09	33.07	30.28	83.8	18.02	68.73
MuLim (kNm)	102	102	102	102	102	102
R	0.606	1.05	0.962	2.661	0.572	2.183
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.231	0.211	0.649	0.2	0.515
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.266	0.266	0.266	0.709	0.709	0.709
AstCalc (sqmm)	170.2	196.87	179.4	552.66	170.2	438.09
AstPrv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement Provided	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	0.709	0.709	0.709
Vu (kN)	64.44	34.48	57.52
Tu (kNm)	0.55	0.55	0.15
V_{Tu} (kN)	3.8	3.8	1.05
Vut (kN)	68.24	38.28	58.57
Tv (N/sqmm)	0.8	0.45	0.69
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85
Vus=Vut-Vc (kN)	20.39	0	10.72
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	229.408	135.65	187.705
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G7**
Beam No : **B20**
Analysis Reference (Member) : 190 (1.981 m)
Beam Length : 2134 mm
Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456
Beam Type : Cantilever Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	14	12	12	12	5	5
Critical L/C - RCDC	10	8	8	8	1	1
Mu (kNm)	3.16	-	-	12.97	69.72	97.75
Tu (kNm)	5.27	5.67	5.67	5.67	0	0
M_{Tu} (kNm)	8.89	9.57	9.57	9.57	0	0
Mud (kNm)	12.06	9.57	9.57	22.54	69.72	97.75
MuLim (kNm)	102	102	102	102	102	102
R	0.383	0.304	0.304	0.716	2.214	3.104
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.523	0.785

Pcclc (%) (C)	0.038	0.064	0.064	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	170.2	445.32	668.04
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	44.73	50.97	52.97
Tu (kNm)	0	0	0
V_{Tu} (kN)	0	0	0
Vut (kN)	44.73	50.97	52.97
Tv (N/sqmm)	0.53	0.6	0.62
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	-	-
Asv Torsion Prv (sqmm)	-	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B21**	
Analysis Reference (Member)	:	106 (1.981 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	8	13	15
Critical L/C - RCDC	16	9	14	4	9	11
Mu (kNm)	41.49	30.43	48.34	82.87	32.32	62.71
Tu (kNm)	0.16	0.16	0.16	0.12	0.16	0.16
M_{Tu} (kNm)	0.27	0.27	0.27	0.21	0.27	0.27
Mud (kNm)	41.76	30.7	48.61	83.07	32.59	62.99
MuLim (kNm)	102	102	102	102	102	102
R	1.326	0.975	1.544	2.638	1.035	2
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2

Ptclc (%) (B')	0.297	0.214	0.35	0.643	0.228	0.466
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	252.43	181.98	297.55	546.96	193.82	396.63
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	55.1	37.81	46.26
Tu (kNm)	0.12	0.12	0.13
V_{Tu} (kN)	0.86	0.86	0.93
Vut (kN)	55.96	38.67	47.18
Tv (N/sqmm)	0.66	0.45	0.55
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	180.927	126.186	153.444
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B22**	
Analysis Reference (Member)	:	94 (1.981 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	8	18	10
Critical L/C - RCDC	16	9	14	4	14	6
Mu (kNm)	30.48	25.56	33.97	72.4	22.85	59.14
Tu (kNm)	0	0	0	0	0	0
M_{Tu} (kNm)	0	0	0	0	0	0
Mud (kNm)	30.48	25.56	33.97	72.4	22.85	59.14
MuLim (kNm)	102	102	102	102	102	102
R	0.968	0.812	1.079	2.299	0.726	1.878

Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.212	0.2	0.238	0.547	0.2	0.434
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	180.62	170.2	202.52	465.13	170.2	369.54
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	49.01	29.77	44.41
Tu (kNm)	0	0	0
V_{Tu} (kN)	0	0	0
Vut (kN)	49.01	29.77	44.41
Tv (N/sqmm)	0.58	0.35	0.52
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	-	-
Asv Torsion Prv (sqmm)	-	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B23**	
Analysis Reference (Member)	:	82 (1.981 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	8	13	15
Critical L/C - RCDC	16	9	14	4	9	11
Mu (kNm)	28.94	31.73	41.45	72.92	21.65	57.62
Tu (kNm)	0	0.19	0.16	0.19	0.19	0
M_{Tu} (kNm)	0	0.33	0.27	0.32	0.33	0
Mud (kNm)	28.94	32.06	41.72	73.24	21.98	57.62
MuLim (kNm)	102	102	102	102	102	102

R	0.919	1.018	1.325	2.326	0.698	1.83
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.201	0.224	0.296	0.554	0.2	0.422
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	171.04	190.53	252.15	471.44	170.2	358.93
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	50.73	31.67	43.13
Tu (kNm)	0.19	0.19	0
V_{Tu} (kN)	1.31	1.31	0
Vut (kN)	52.04	32.98	43.13
Tv (N/sqmm)	0.61	0.39	0.51
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	170.537	110.203	-
Asv Torsion Prv (sqmm)	365.6	365.6	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G8**	
Beam No	:	**B24**	
Analysis Reference (Member)	:	192 (1.981 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	12	12	12	7	7	5
Critical L/C - RCDC	8	8	8	3	3	1
Mu (kNm)	9.28	-	-	9.98	52.74	82.67
Tu (kNm)	7.36	7.36	7.36	7.12	7.12	3.65
M_{Tu} (kNm)	12.43	12.43	12.43	12.02	12.02	6.16
Mud (kNm)	21.71	12.43	12.43	22	64.76	88.83

MuLim (kNm)	102	102	102	102	102	102
R	0.69	0.395	0.395	0.699	2.057	2.821
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.481	0.697
Pcclc (%) (C)	0.021	0.084	0.084	0.014	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	170.2	409.29	593.15
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	7	7	7
Critical L/C - RCDC	3	3	3
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	37.66	42.5	44.52
Tu (kNm)	7.12	7.12	7.12
V_{Tu} (kN)	49.53	49.53	49.53
Vut (kN)	87.19	92.04	94.05
Tv (N/sqmm)	1.02	1.08	1.11
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	30.5	35.35	37.36
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	494.846	510.172	516.547
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	264.62	279.7
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140

Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G8**	
Beam No	:	**B25**	
Analysis Reference (Member)	:	107 (1.981 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design						
	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**

Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11
Mu (kNm)	33.71	31.55	38.64	96.13	30.83	78.7
Tu (kNm)	0.49	0.31	0.11	0.31	0.31	0.69
M_{Tu} (kNm)	0.83	0.52	0.18	0.52	0.52	1.17
Mud (kNm)	34.54	32.07	38.82	96.65	31.34	79.87
MuLim (kNm)	102	102	102	102	102	102
R	1.097	1.018	1.233	3.069	0.995	2.537
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.242	0.224	0.274	0.774	0.219	0.613
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	206.14	190.56	233.42	658.56	186.04	521.87
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	68.24	41.25	62.02
Tu (kNm)	0.31	0.31	0.69
V_{Tu} (kN)	2.13	2.13	4.81
Vut (kN)	70.37	43.38	66.83
Tv (N/sqmm)	0.83	0.51	0.79
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	13.68	0	10.14
Legs	2	2	2

Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	232.155	146.736	232.798
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G8**	
Beam No	:	**B26**	
Analysis Reference (Member)	:	95 (1.981 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm

Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	18	15
Critical L/C - RCDC	16	9	14	9	14	11
Mu (kNm)	25.64	30.84	29.38	89.12	21.44	80.65
Tu (kNm)	0.13	0.53	0.43	0.53	0.43	0
M_{Tu} (kNm)	0.21	0.89	0.72	0.89	0.72	0
Mud (kNm)	25.85	31.73	30.1	90.01	22.16	80.65
MuLim (kNm)	102	102	102	102	102	102
R	0.821	1.008	0.956	2.859	0.704	2.561
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.221	0.209	0.708	0.2	0.62
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	170.2	188.43	178.26	602.85	170.2	527.95
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	8	13	15
Critical L/C - RCDC	4	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	66.08	35.38	62.35
Tu (kNm)	0.5	0.53	0
	3.45	3.67	0

V_{Tu} (kN)			
Vut (kN)	69.53	39.05	62.35
Tv (N/sqmm)	0.82	0.46	0.73
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	12.84	0	5.66
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	235.326	139.814	-
Asv Torsion Prv (sqmm)	718.14	365.6	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G8**	
Beam No	:	**B27**	
Analysis Reference (Member)	:	83 (1.981 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	

Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	13	13	18	15
Critical L/C - RCDC	16	9	9	9	14	11
Mu (kNm)	26.65	39.11	38.42	94.85	22.98	80.35
Tu (kNm)	0.68	0.99	0.99	0.99	0.58	1.1
M_{Tu} (kNm)	1.15	1.68	1.68	1.68	0.97	1.86
Mud (kNm)	27.8	40.79	40.1	96.53	23.95	82.21
MuLim (kNm)	102	102	102	102	102	102
R	0.883	1.295	1.274	3.066	0.761	2.611
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.289	0.284	0.773	0.2	0.635
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	170.2	246.13	241.68	657.54	170.2	540.17
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	69.5	39.54	62.18
Tu (kNm)	1	1	1.1
V_{Tu} (kN)	6.92	6.92	7.66
Vut (kN)	76.42	46.46	69.84
Tv (N/sqmm)	0.9	0.55	0.82
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	19.73	0	13.15
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	272.467	177.637	254.933
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group : **G9**

Beam No : **B28**

Analysis Reference (Member)	:	199 (1.981 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	7	7	7	7	8	8
Critical L/C - RCDC	3	3	3	3	4	4
Mu (kNm)	-	-	-	9.88	24.3	34.96
Tu (kNm)	8.3	8.3	8.3	8.3	7.66	7.66
M_{Tu} (kNm)	14.01	14.01	14.01	14.01	12.93	12.93
Mud (kNm)	14.01	14.01	14.01	23.89	37.22	47.89
MuLim (kNm)	102	102	102	102	102	102
R	0.445	0.445	0.445	0.759	1.182	1.521
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.262	0.344
Pcclc (%) (C)	0.095	0.095	0.095	0.028	0	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	223.2	292.74
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural

				reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	8	8	5
Critical L/C - RCDC	4	4	1
PtPrv (%)	0.399	0.399	0.399
Vu (kN)	9.72	16.34	22.64
Tu (kNm)	7.66	7.66	7.45
V_{Tu} (kN)	53.28	53.28	51.81
Vut (kN)	63	69.61	74.45
Tv (N/sqmm)	0.74	0.82	0.87
Tc (N/sqmm)	0.45	0.45	0.45
Vc (kN)	37.88	37.88	37.88
Vus=Vut-Vc (kN)	25.12	31.73	36.57
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	405.096	425.56	434.726
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	273.69
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B29**	
Analysis Reference (Member)	:	200 (1.981 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	15	12	8	13	15
Critical L/C - RCDC	16	11	8	4	9	11
Mu (kNm)	7.22	7.45	0.73	38.89	16.02	17.18
Tu (kNm)	1.11	1.4	1.44	0.12	0	1.4
M_{Tu} (kNm)	1.87	2.36	2.43	0.21	0	2.36
Mud (kNm)	9.09	9.81	3.16	39.1	16.02	19.54
MuLim (kNm)	102	102	102	102	102	102
R	0.289	0.312	0.1	1.242	0.509	0.621
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.276	0.2	0.2
Pcclc (%) (C)	0	0	0.011	0	0	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	235.2	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	8	15
Critical L/C - RCDC	1	4	11
PtPrv (%)	0.399	0.399	0.399
Vu (kN)	26.49	13.21	19.23
Tu (kNm)	0.83	0.12	1.4
V_{Tu} (kN)	5.79	0.86	9.73
Vut (kN)	32.28	14.07	28.97
Tv (N/sqmm)	0.38	0.17	0.34
Tc (N/sqmm)	0.45	0.45	0.45
Vc (kN)	37.88	37.88	37.88
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	122.751	46.937	128.023
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm

Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm
Side Face Reinforcement not required.		

Group	:	**G9**	
Beam No	:	**B30**	
Analysis Reference (Member)	:	201 (1.981 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	5	5	13	5	13	5
Critical L/C - RCDC	1	1	9	1	9	1
Mu (kNm)	5.52	13.31	5.14	19.03	-	21.83
Tu (kNm)	0.73	0.73	0.93	0.73	0.93	0.73
M_{Tu} (kNm)	1.24	1.24	1.57	1.24	1.57	1.24
Mud (kNm)	6.75	14.55	6.71	20.26	1.57	23.07
MuLim (kNm)	102	102	102	102	102	102
R	0.214	0.462	0.213	0.644	0.05	0.733
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.01	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	0.399	0.266	0.399
Vu (kN)	25.22	9.5	26.33
Tu (kNm)	0.73	0.73	0.73
V_{Tu} (kN)	5.09	5.09	5.09
Vut (kN)	30.31	14.59	31.42
Tv (N/sqmm)	0.36	0.17	0.37
Tc (N/sqmm)	0.45	0.37	0.45
Vc (kN)	37.88	31.88	37.88
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	113.894	65.239	117.323
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B31**	
Analysis Reference (Member)	:	202 (1.981 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	5	5	8	5	5	15
Critical L/C - RCDC	1	1	4	1	1	11
Mu (kNm)	5.87	19.44	16.56	23.3	-	12.35
Tu (kNm)	2.65	2.65	2.23	2.65	2.65	1.82
M_{Tu} (kNm)	4.47	4.47	3.76	4.47	4.47	3.07
Mud (kNm)	10.34	23.91	20.32	27.77	4.47	15.42
MuLim (kNm)	102	102	102	102	102	102
R	0.329	0.759	0.645	0.882	0.142	0.49
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.03	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	0.399	0.266	0.399
Vu (kN)	29.12	12.56	21.62
Tu (kNm)	2.65	2.65	2.65
V_{Tu} (kN)	18.42	18.42	18.42
Vut (kN)	47.55	30.99	40.04
Tv (N/sqmm)	0.56	0.36	0.47
Tc (N/sqmm)	0.45	0.37	0.45
Vc (kN)	37.88	31.88	37.88
Vus=Vut-Vc (kN)	9.67	0	2.16
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	219.821	168.569	196.591
Asv Torsion Prv (sqmm)	718.14	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	140
SvPrv (mm)	140	275	140
Asv Total Prv (sqmm)	718.14	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.
